

## Project Title

### Serverless Event-Driven Architecture using AWS Lambda and SNS

## Objective

To design and implement a **serverless event-driven system** where a message published to an SNS topic automatically triggers a Lambda function, which processes the message and logs the output.

## Services Used

- **AWS Lambda** – to execute code without managing servers
- **Amazon Simple Notification Service** – to send messages and trigger events
- **Amazon CloudWatch** – to monitor logs and execution
- **IAM** – to manage permissions

## Architecture

SNS Topic → Lambda Function → CloudWatch Logs

## Implementation Steps

### Step 1: Create Lambda Function

1. Open AWS Management Console
2. Navigate to **AWS Lambda**
3. Click **Create function**
4. Select **Author from scratch**
5. Provide the following details:
  - Function Name: lambda-sns-project
  - Runtime: Python 3.12
  - Permissions: Create new role with basic Lambda permissions
6. Click **Create function**

## Write Lambda Function Code

Replace the default code with:

```
import json
```

```
def lambda_handler(event, context):
```

```
    print("SNS Event Received")
```

```
    for record in event['Records']:
```

```
        message = record['Sns']['Message']
```

```
        print("Message from SNS:", message)
```

```
    return {
```

```
        'statusCode': 200,
```

```
        'body': json.dumps('Message processed successfully')
```

```
    }
```

Click Deploy

### Create SNS Topic

1. Navigate to **Amazon SNS**
2. Click **Create topic**
3. Select **FIFO Topic**
4. Enter topic name: lambda-sns-topic.fifo
5. Click **Create topic**

Create Subscription

Open the created SNS topic

Click Create subscription

Select protocol: AWS Lambda

Choose endpoint: lambda-sns-project

Click Create subscription

### **Publish Message to SNS**

1. Go to SNS topic
2. Click **Publish message**

Since it is a FIFO topic, provide:

- Message Group ID: group1
  - Message Deduplication ID: msg1
3. Enter message:

Hello from SNS

4. Click **Publish message**

Trigger and Execution Flow:

SNS receives the message

SNS triggers Lambda automatically

Lambda processes the message

Output is stored in CloudWatch logs

### **Verify Output in CloudWatch**

1. Go to **Amazon CloudWatch**
2. Navigate to **Log groups**
3. Open:

/aws/lambda/lambda-sns-project

4. Open latest log stream

Expected output:

SNS Event Received

Message from SNS: Hello from SNS

Issues Faced and Resolutions

Issue 1: KeyError: 'Records'

Cause:

Lambda test event did not contain SNS event structure

Solution:

Used proper SNS event format or triggered via SNS

Issue 2: FIFO Topic Publishing Error

Error:

Message Group ID must not be empty

Deduplication ID must not be empty

Cause:

FIFO topics require additional parameters

Solution:

Provided:

Message Group ID → group1

Deduplication ID → msg1

Issue 3: Lambda Not Triggering (Common Scenario)

Cause:

SNS subscription not configured properly

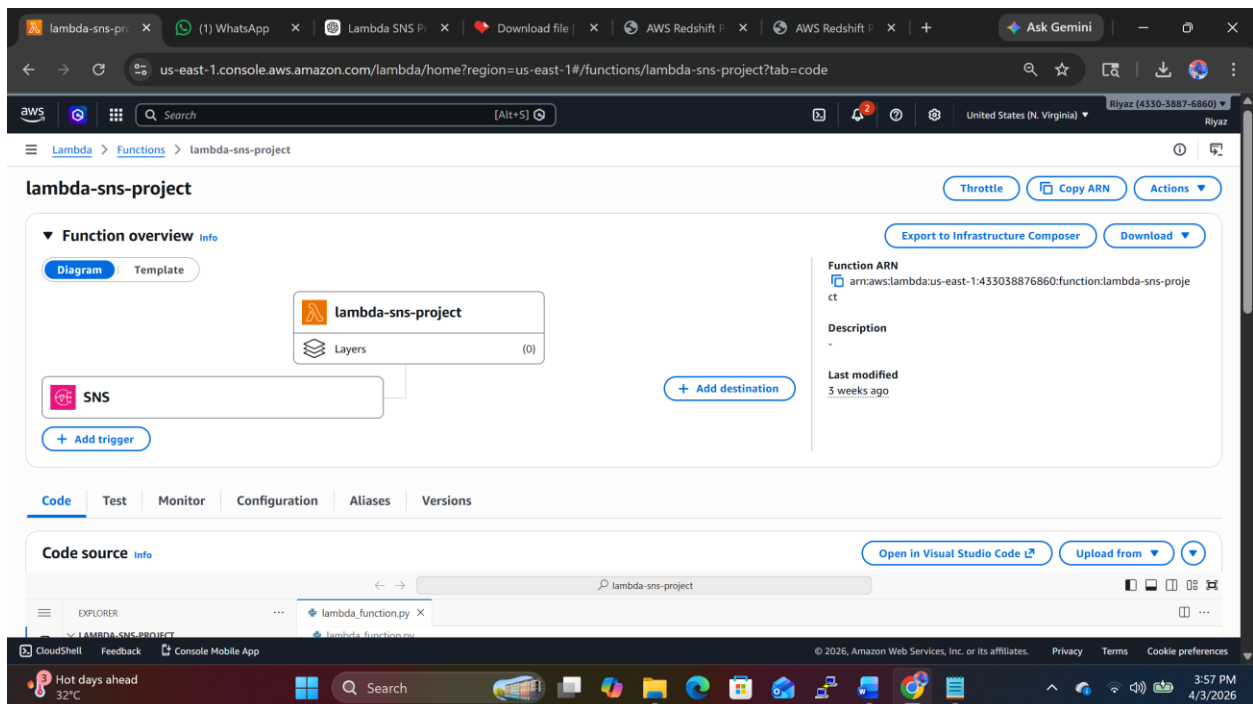
Solution:

Created subscription with protocol AWS Lambda

## ◆ Outcome

Successfully implemented a serverless event-driven architecture where SNS triggers Lambda to process messages in real time, with logs monitored using CloudWatch.

Lambda output:



## Cloudwatch log group:

The screenshot displays the AWS CloudWatch console interface. The browser address bar shows the URL: `us-east-1.console.aws.amazon.com/cloudwatch/home?region=us-east-1#logsV2:log-groups/log-group/$252Faws$252Fiambda$252Fiambd...`. The console header includes the AWS logo, a search bar with "cloudwatch", and navigation links for "Log management" and "aws/lambda/lambda-sns-project". The left sidebar contains a "CloudWatch" menu with sections for "Logs", "Metrics", and "Network Monitoring". The "Logs" section is expanded, showing "Log Management", "Log Anomalies", "Live Tail", "Logs Insights", and "Contributor Insights". The "Metrics" section shows "All metrics", "Explorer", and "Streams". The "Network Monitoring" section shows "Flow monitors", "Internet monitors", and "Synthetic monitors". The main content area is titled "Log events" and includes a search bar with the text "Filter events - press enter to search". Below the search bar is a "Display" button. The log events are listed in a table with columns for "Timestamp" and "Message". The messages include "INIT\_START Runtime Version: python:3.12.mainlinev2.v3 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:27990c3d0965bea97240f33808b8d623b69...", "START RequestId: 3f9321d7-43b4-459f-beda-ed2463273656 Version: \$LATEST", "SNS Event Received", "Message from SNS: Hello from SNS", "END RequestId: 3f9321d7-43b4-459f-beda-ed2463273656", and "REPORT RequestId: 3f9321d7-43b4-459f-beda-ed2463273656 Duration: 2.00 ms Billed Duration: 101 ms Memory Size: 128 MB Max Memory Used: 36 MB Ini...". The bottom of the screenshot shows the Windows taskbar with the date and time "4:09 PM 4/3/2026".

CloudWatch

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear 1m 30m 1h 12h Custom UTC timezone

Display

| Timestamp                | Message                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | There are older events to load. <a href="#">Load more</a> .                                                                                         |
| 2026-03-12T04:34:15.168Z | INIT_START Runtime Version: python:3.12.mainlinev2.v3 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:27990c3d0965bea97240f33808b8d623b69... |
| 2026-03-12T04:34:15.278Z | START RequestId: 3f9321d7-43b4-459f-beda-ed2463273656 Version: \$LATEST                                                                             |
| 2026-03-12T04:34:15.271Z | SNS Event Received                                                                                                                                  |
| 2026-03-12T04:34:15.271Z | Message from SNS: Hello from SNS                                                                                                                    |
| 2026-03-12T04:34:15.273Z | END RequestId: 3f9321d7-43b4-459f-beda-ed2463273656                                                                                                 |
| 2026-03-12T04:34:15.273Z | REPORT RequestId: 3f9321d7-43b4-459f-beda-ed2463273656 Duration: 2.00 ms Billed Duration: 101 ms Memory Size: 128 MB Max Memory Used: 36 MB Ini...  |
|                          | No newer events at this moment. Auto retry paused. <a href="#">Resume</a>                                                                           |